

FRAME

Memories shouldn't be work!

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problem statement, value proposition & solution



TASKS

simple, moderate & complex



RESULTS

usability goals & key measurements



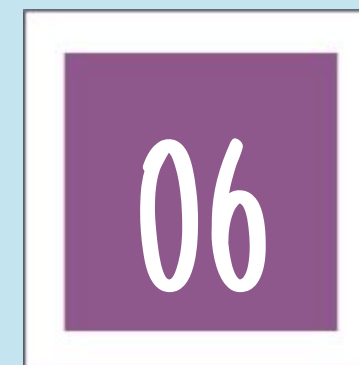
REVISED INTERFACES

changed simple, moderate & complex sketches

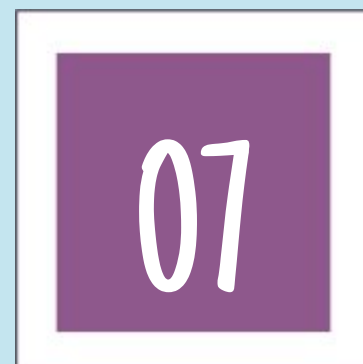
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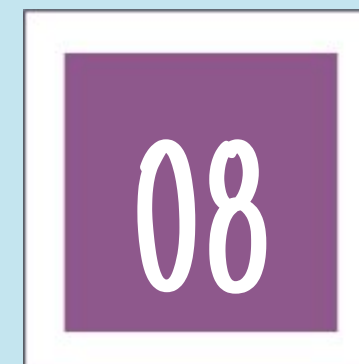
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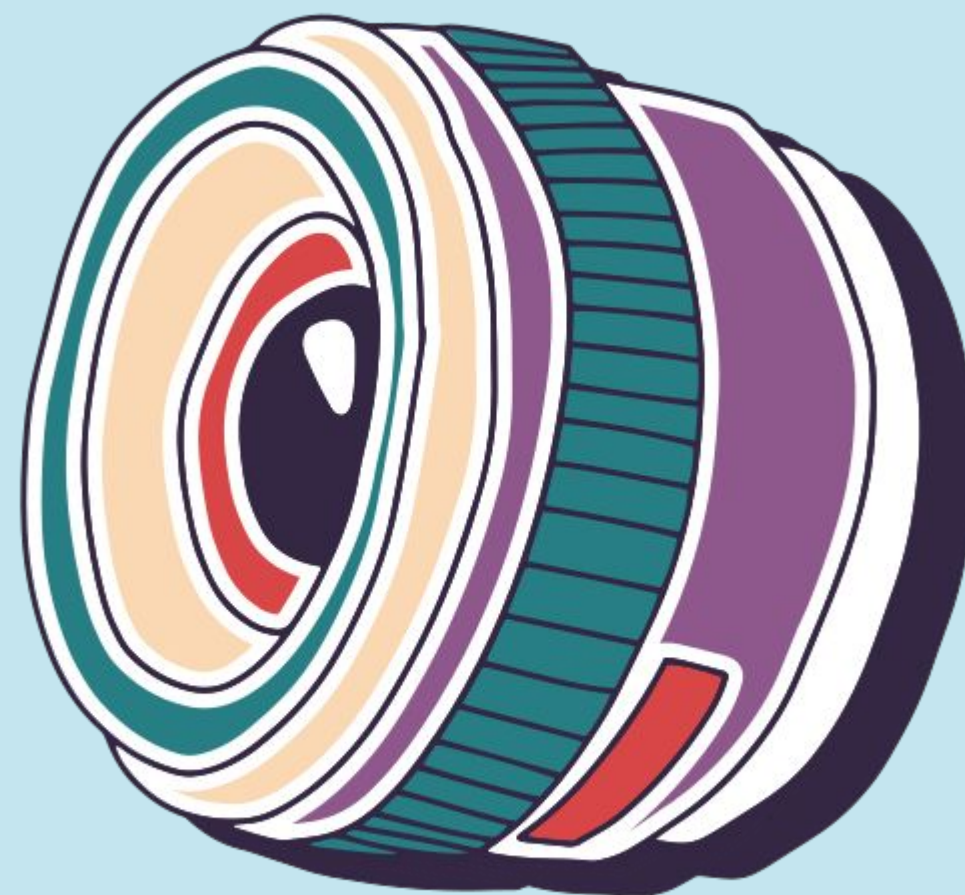
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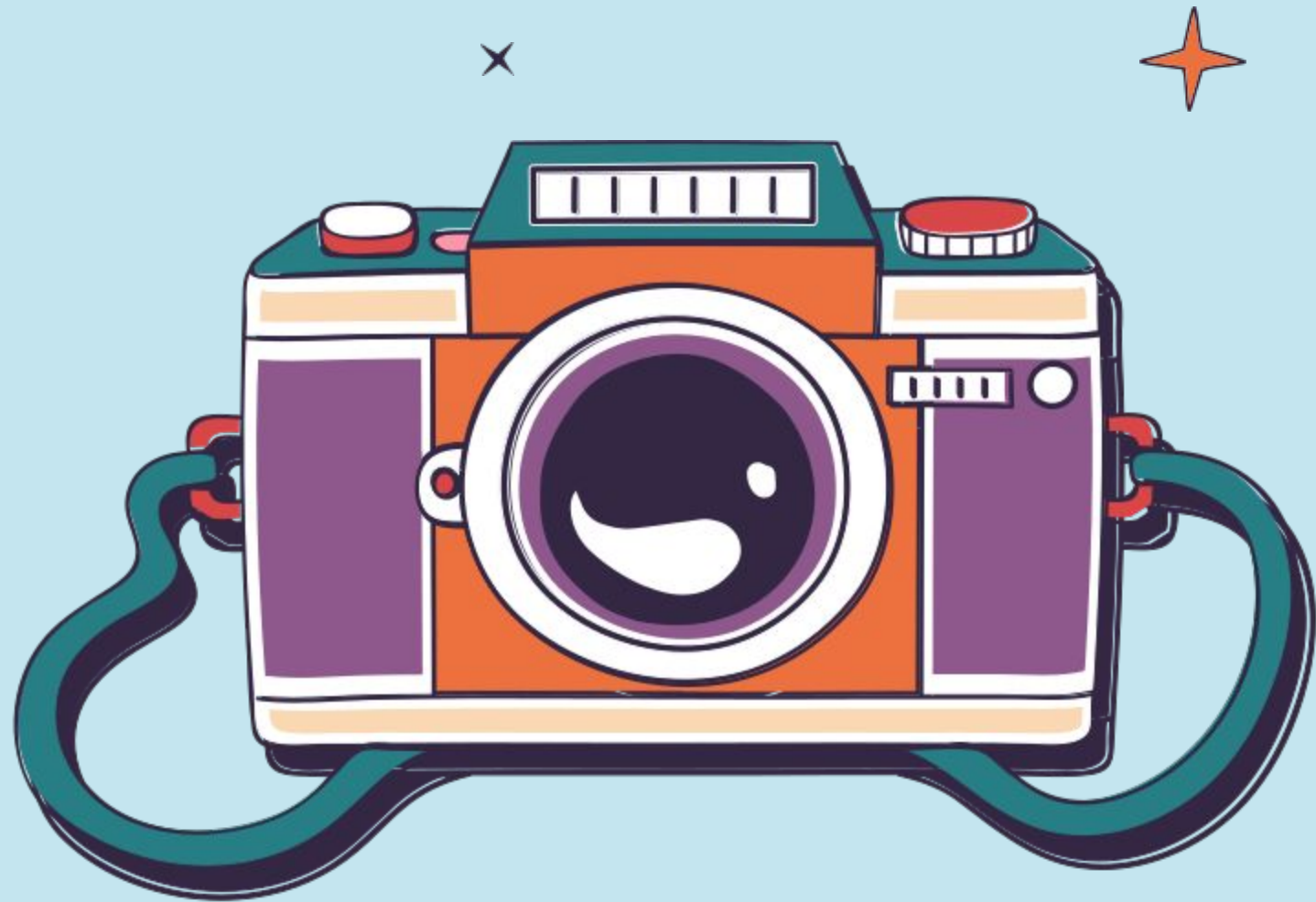
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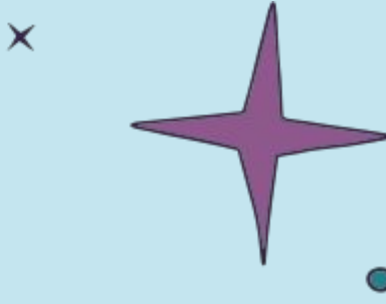
intro



problem statement



- Photo sharing breaks down after the moment, turning memories into manual labor involving sorting, app-switching, and follow-ups instead of enjoyment.
- People lose confidence once photos are shared, worrying about who has access, how long they'll have it, and whether boundaries are respected over time.
- The burden falls on one "default sharer", who becomes a human bridge across iOS and Android, sacrificing time and mental energy to keep everyone included.



FRAME SHARING MEMORIES MADE



Value Proposition Statement SIMPLE

Frame transforms scattered photos into shared, organized memories by automatically collecting, curating, and synchronizing albums across groups

Solution



Frame turns scattered event photos on iOS or Android into unified shared albums automatically. Frame combines intelligent event clustering, privacy-focused facial detection, and cross-platform synchronization to remove the burden of sorting and ensure memories reach the intended audience effortlessly.



our proposed solution is cross-platform intentional sharing



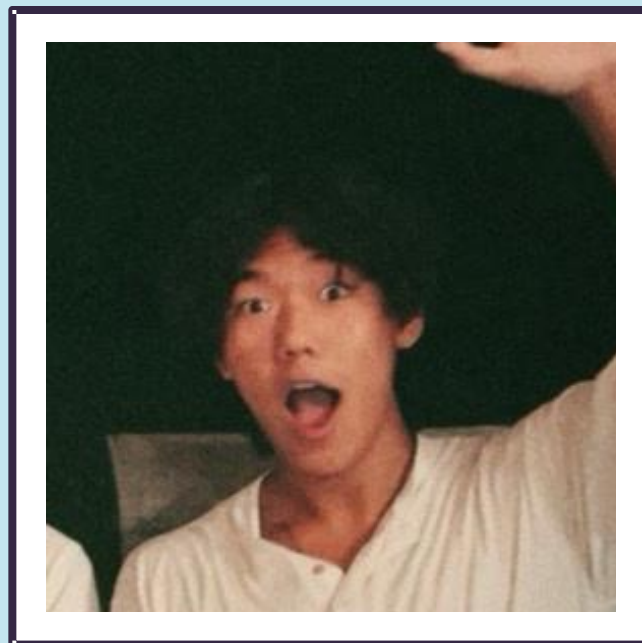


solution

Frame is a cross-platform photo sharing experience designed to keep memories effortless after the moment ends. It removes the burden of manual sorting, app-switching, and follow-ups by automatically creating shared albums and making audience visibility intentional and clear. Instead of turning one person into the “default sharer,” Frame helps groups share photos seamlessly across

iOS and Android without losing track, time, or momentum.

The people behind FRAME



GAVIN CAO

loves animals, food & nature
lowkey a farmer :)



**RACHIT
GUPTA**

loves to drive, travel, and
click pictures!!

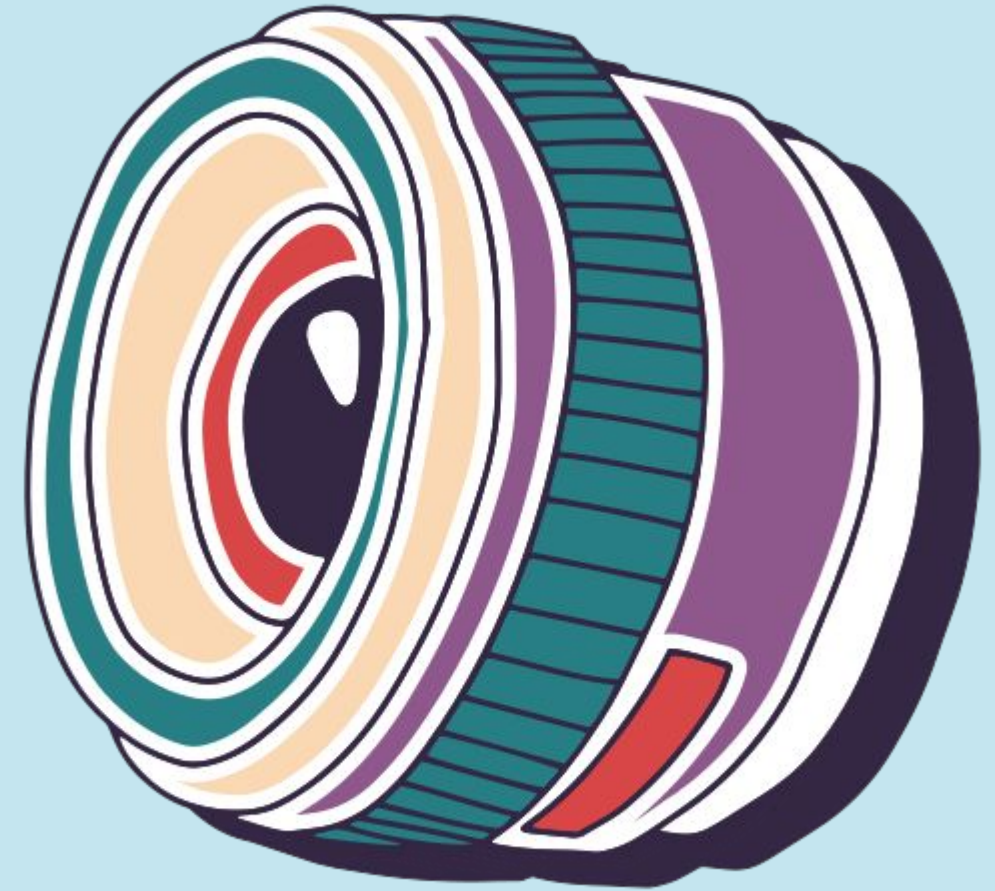


ZEEL PATEL

loves to sleep, eat
and repeat

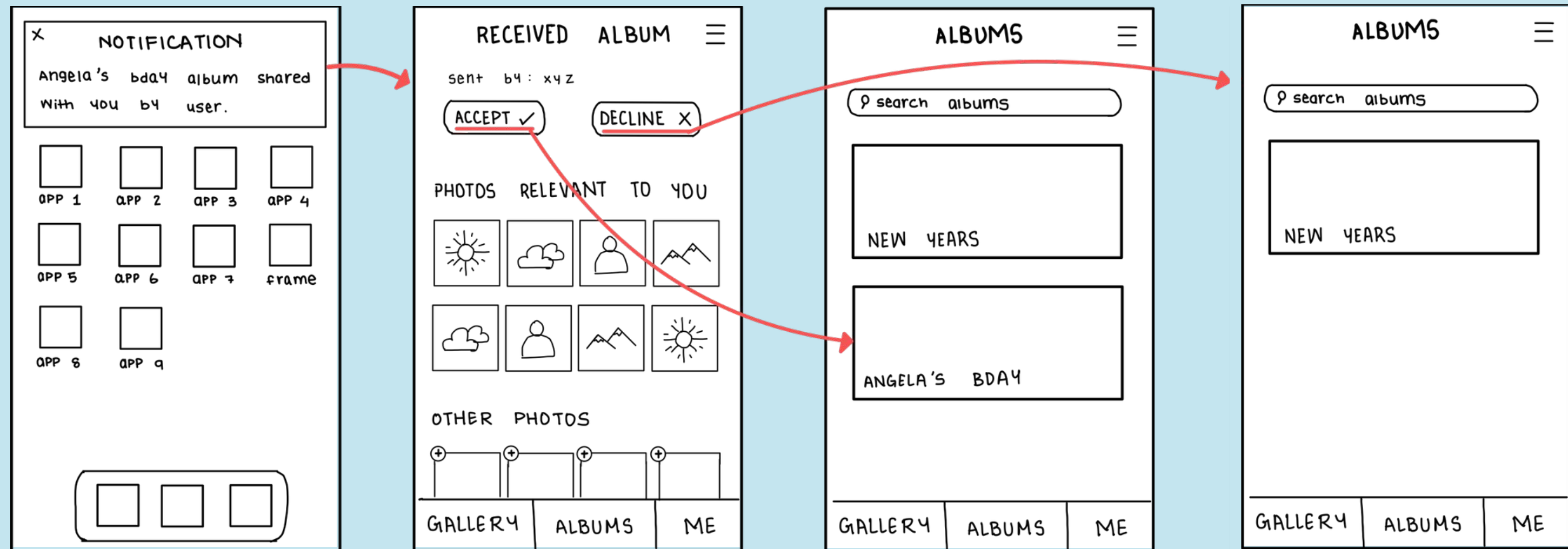


tasks



simple

User receives shared album and accepts or decline it



gets a notification that an album is shared to the user

user can view who sent it, all the most relevant photos and other photos of the album and then accept or decline

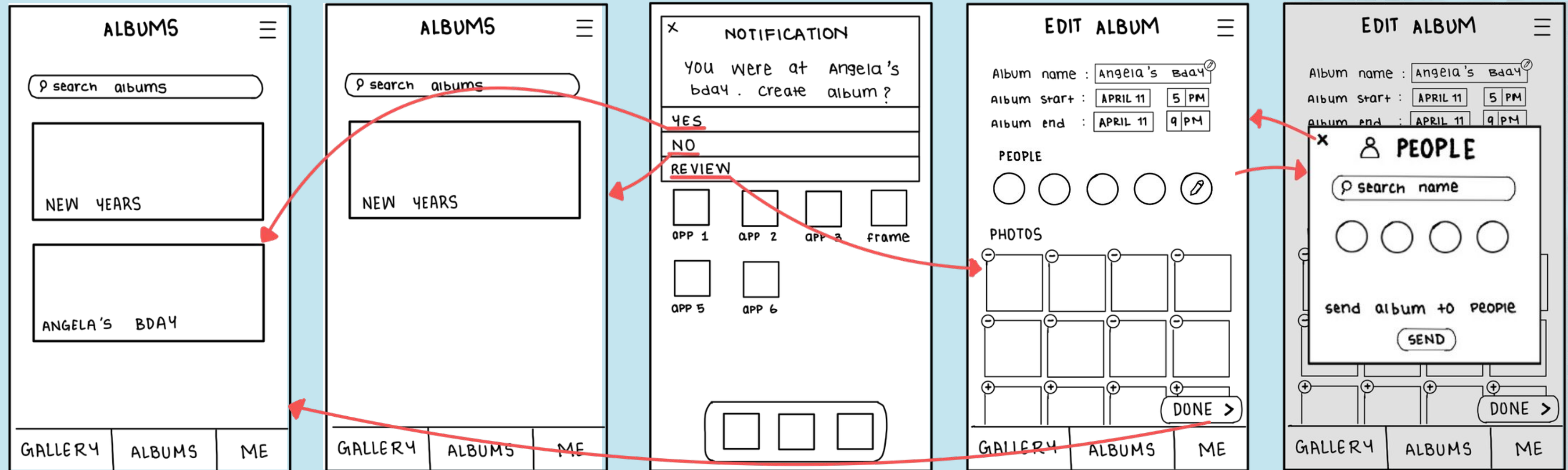
if accepted, the shared album is in the album screen

if declined, the old albums are in the album screen

Reasoning: This is simple from the user's perspective because it's a quick, low-effort decision flow: the user gets a clear notification, previews what the album is, and taps Accept or Decline. The outcome is immediate (album appears in Albums if accepted; nothing changes if declined)

moderate

User discovers automatically curated albums by frame



if yes, the new albums is added to the album screen

if no, the old albums are in the album screen

user gets a notification once an album is automatically created, and they have options to choose from for the album

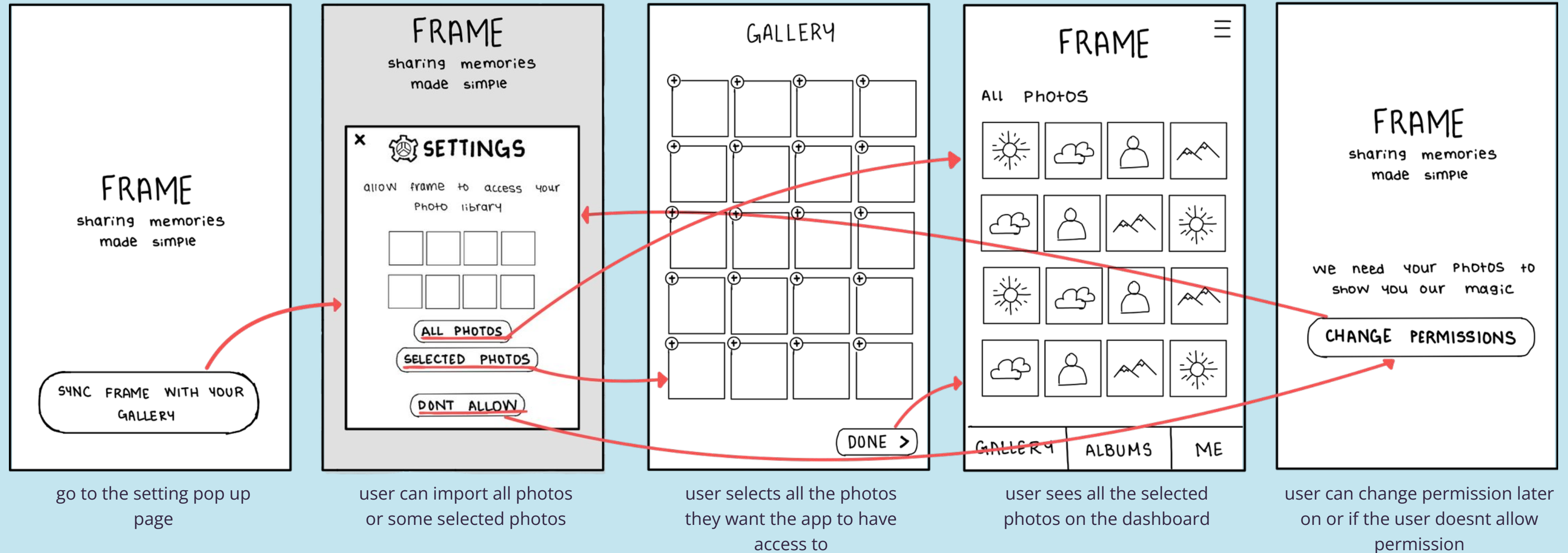
user can review and edit the album by adding or removing photos, changing the timeline and information like the name of the album

user can click and edit information of the timelines and people

Reasoning: This is moderate because the user are asked to evaluate an automatically created album, then possibly review photos, adjust details (name/timeline), and manage people before finishing. The steps aren't hard, but they require more attention and increases effort.

complex

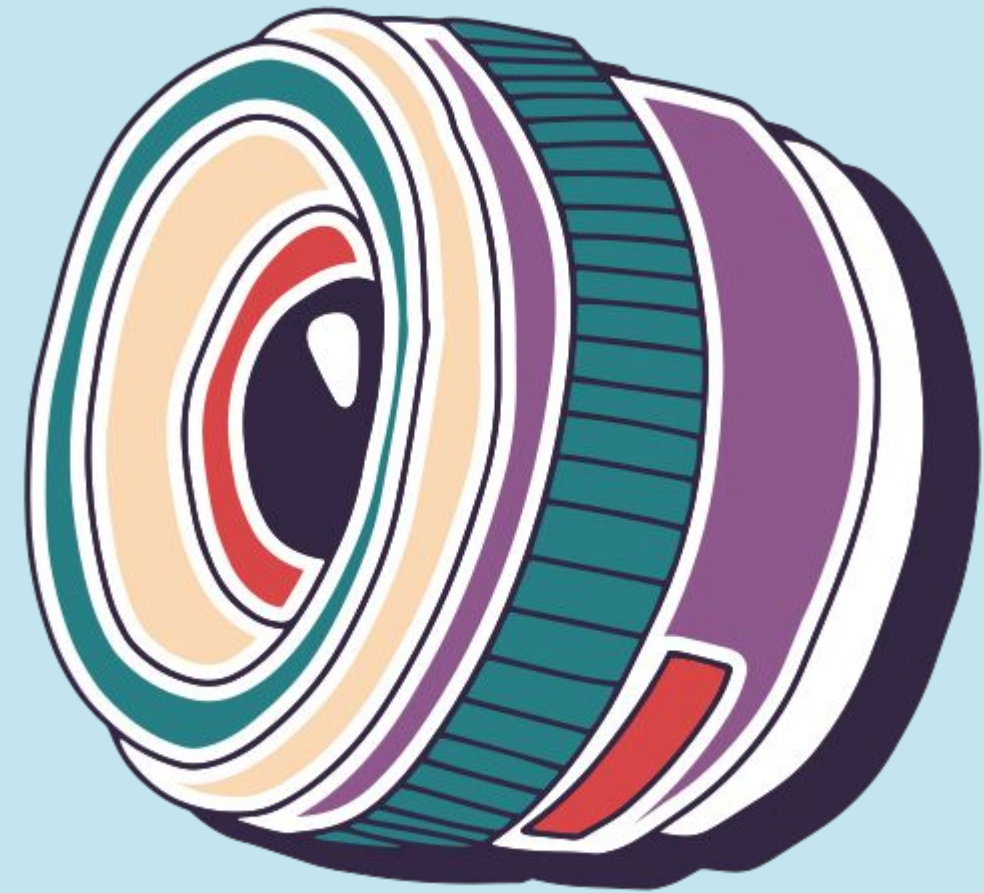
User imports all or selected photos from his gallery into the app



Reasoning: This is complex because it involves setup + privacy choices + selection work: the user must understand a permission prompt, decide between all photos vs selected, potentially scroll/select many items, and then confirm access. From the user side, this is higher stakes (privacy), and more time/attention, which adds complexity.



results



Usability Goals & Measurements

Goal 1: Learnability (Intuitive Navigation)

- **Rationale:** Frame introduces a new mental model (auto-created albums + auto-sharing) Users must quickly understand how to control who is included and what is shared.
- **Key Measurement:** Navigation Errors (# misclicks per task), Second-Attempt Errors (# errors when repeating task)
- **Why?** If users fail once but succeed without confusion on the second attempt, the issue may be surface-level UI clarity rather than structural flow problems.



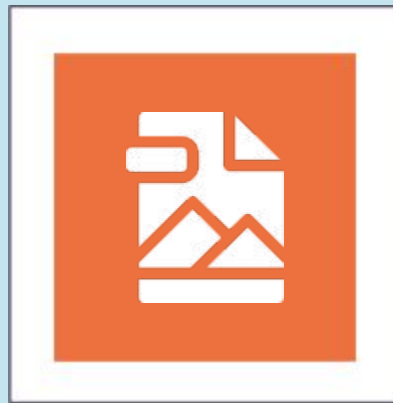
Goal 2: Efficiency (Task Completion Speed Within Frame)

- **Rationale:** Frame's value proposition is reducing manual effort. Core tasks (accepting, editing, and sharing albums) should require minimal steps and minimal time within the app.
- **Key Measurement:** Time on Task (seconds), Measured from: Notification → Shared Album Success & Auto-creation → Final "Done"
- **Why?** Time on task directly reflects interaction efficiency. Faster completion (with actual secs) suggests lower friction and better usability.

Collected Data (Metrics)

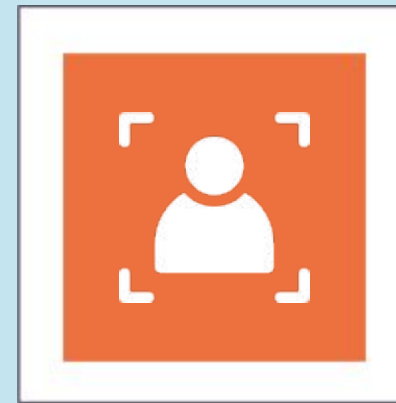
Participant	Misclicks	Task Completion Time	Mental Effort
Participant 1	1 (Edit Flow)	23 s	2 (Low)
Participant 2	3 (Sync Flow)	33 s	4
Participant 3	0	18 s	1

Progress Towards Goals



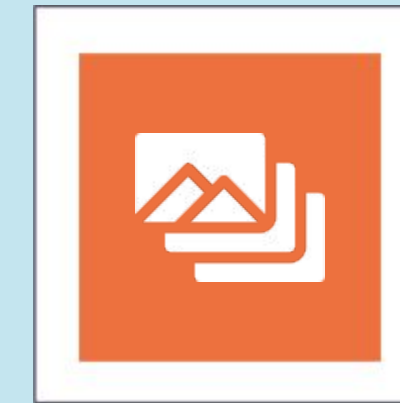
Learnability

Users quickly understood the Accept/Decline flow (0 misclicks, 100% success), suggesting strong initial learnability. However, the Edit People feature caused early navigation errors (avg 2 misclicks), indicating discoverability issues. Since all users corrected their behavior on the second attempt, the structural flow is sound, but visual affordances need improvement.



Efficiency

The simple sharing task averaged 6–8 seconds, showing strong efficiency in core value areas. However, the full auto-creation editing flow averaged 45–60 seconds due to hesitation around permissions and timeline controls. This indicates that while automation reduces manual selection effort, unclear controls reduce speed.



Mental Effort

Users reported low mental effort during simple tasks (1–2/5), but moderate to high effort during permission and timeline editing screens (3–4/5). Confusion around “sync” language and slider controls increased cognitive load. Reducing ambiguity in these screens would lower mental effort and improve overall usability.

Necessary Future Changes

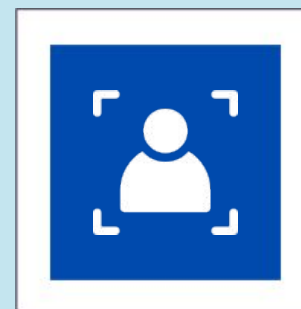


Improve Discoverability of "Edit People"

Change: Make avatars directly tappable to remove/add people instead of requiring the pencil icon.

Why: 2/3 users misclicked before discovering the edit control.

Expected impact: Fewer misclicks + ~5–10 sec reduction in edit flow



Replace Timeline Slider with Clear Pickers

Change: Replace abstract sliders with calendar/time selector UI

Why: All users hesitated during timeline adjustment (20–25 sec range). Sliders increased cognitive load and uncertainty.

Expected impact:

- Reduce hesitation time
- Lower mental effort
- Improve efficiency of moderate task



Clarify Gallery Permission Messaging

Change: Replace vague "Sync Frame with your gallery" with brief explanatory subtext

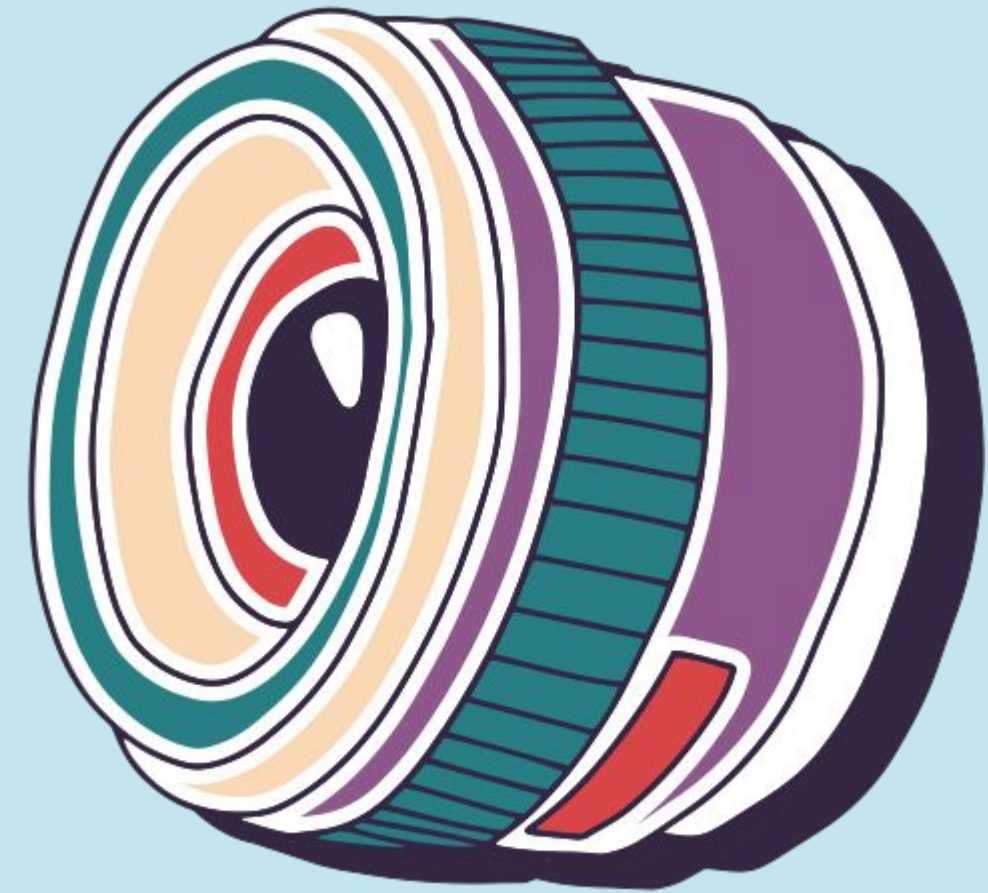
Why: All users hesitated 10–15 sec due to privacy uncertainty. This hesitation increased total completion time

Expected impact:

- Reduce hesitation time
- Lower perceived risk
- Improve efficiency without sacrificing trust



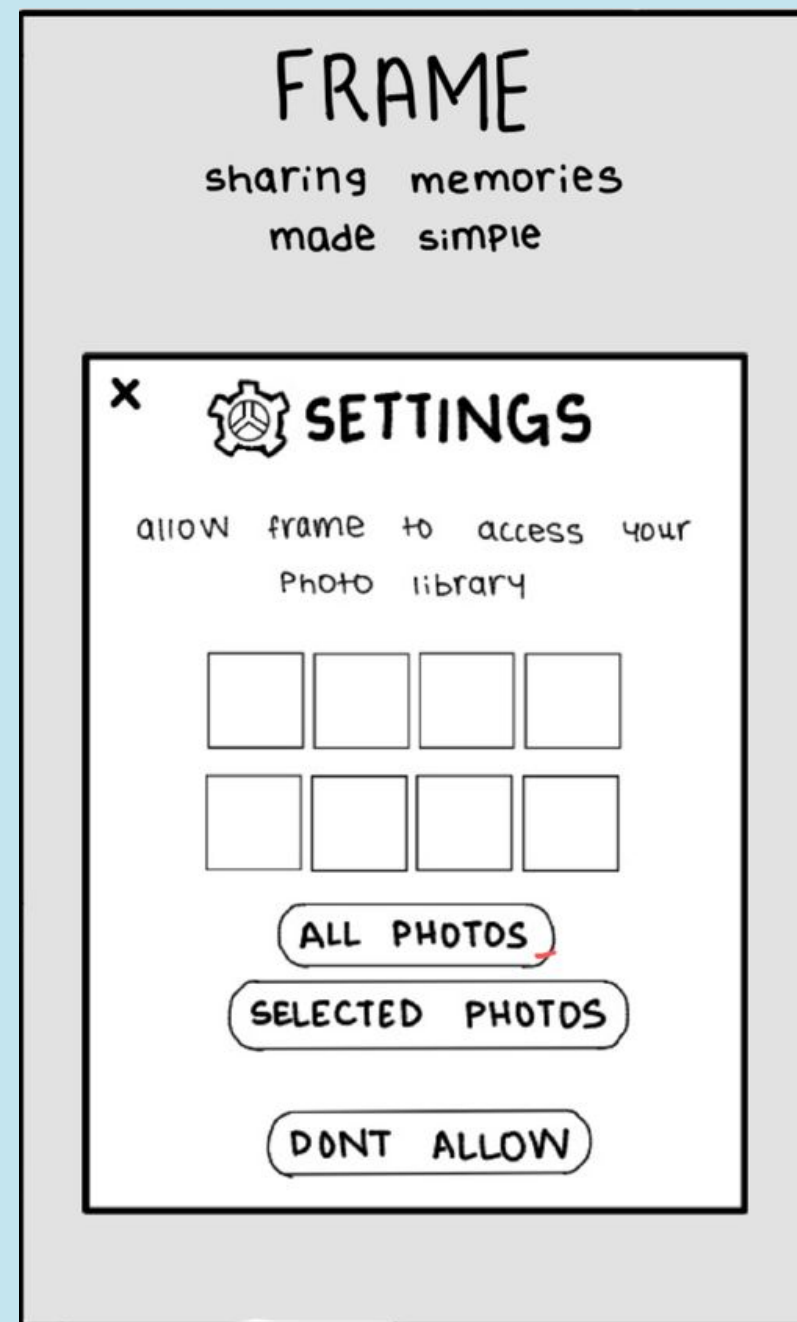
revised interface sketches



Major Change 1 - Resolving Privacy Anxiety

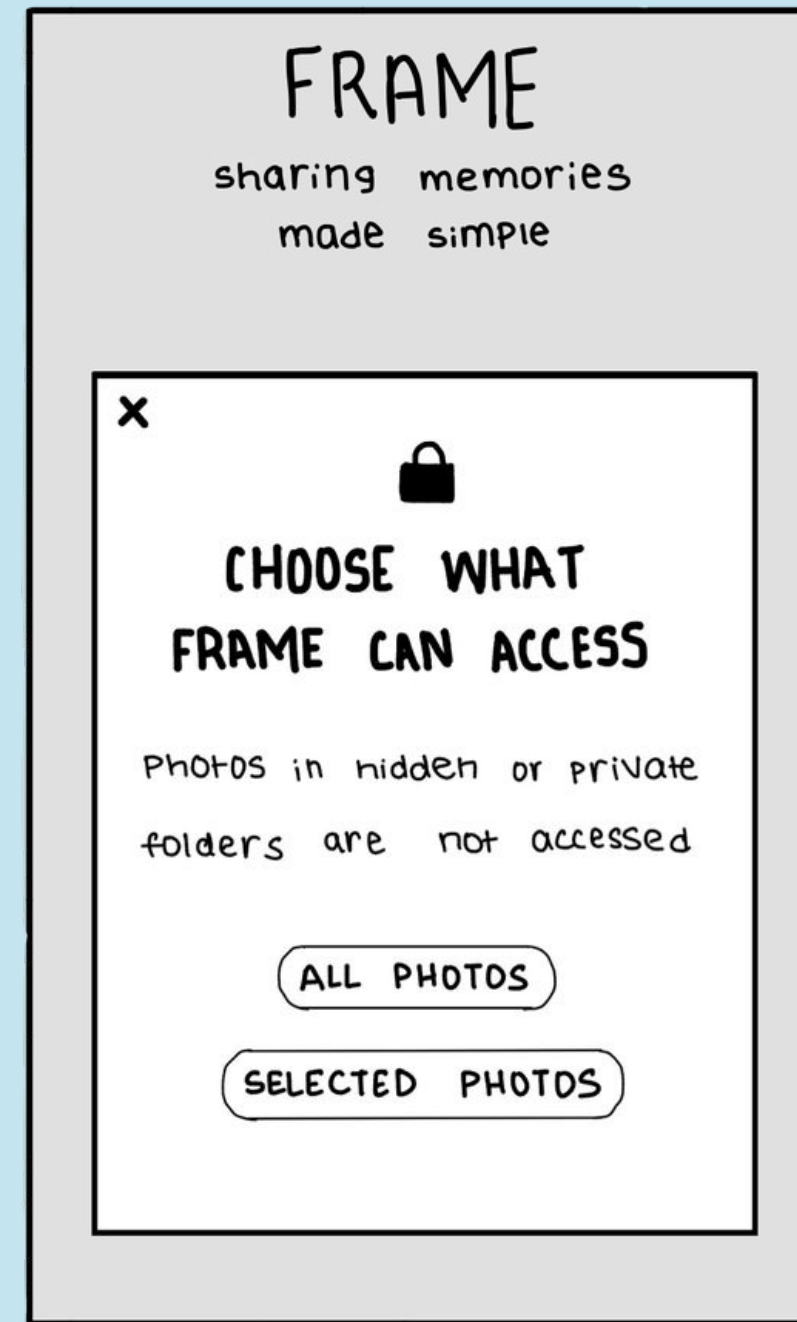
Previously

A generic settings modal with technical "Sync" jargon that caused user hesitation.



Now

We switched our old internal import system to instead be the native OS-style privacy pop up



Benefits of Change #1

Evidence from Testing:

- During low-fi testing, all 3 users hesitated at the gallery permission screen (Severity 4 incident). One user explicitly asked, "Is this going to upload my screenshots too?" Peer feedback strongly recommended clearer permission messaging.

Efficiency (Time on Task):

- By replacing the anxiety-inducing word "Sync" with reassuring, human-centered language, we eliminate user hesitation. When users aren't pausing to worry about their private data, they confidently complete the flow, directly speeding up task completion times.



Major Change 2 - Upgrading to Native Timeline Pickers

Previously

Plain rectangular text boxes for album start/end times that lacked clear interaction affordances.

The previous 'EDIT ALBUM' screen mockup features a title bar 'EDIT ALBUM' with a hamburger menu icon. Below the title bar, there are three text input fields: 'Album name : Angela's Bday', 'Album start : APRIL 11 5 PM', and 'Album end : APRIL 11 9 PM'. The 'Album start' and 'Album end' fields are composed of two separate rectangular boxes for the date and time. Below these fields are two sections: 'PEOPLE' with four empty circles and a pencil icon, and 'PHOTOS' with a 3x4 grid of empty square boxes. Each box in the 'PHOTOS' grid has a small circular icon in its top-left corner (minus for removal, plus for addition). At the bottom right of the 'PHOTOS' grid is a 'SEND >' button. The bottom of the screen has a navigation bar with three tabs: 'GALLERY', 'ALBUMS', and 'ME'.

Now

Standard, native OS-style scrolling date picker that users already know how to operate.

The new 'EDIT ALBUM' screen mockup features a title bar 'EDIT ALBUM' with a hamburger menu icon. Below the title bar, there are two text input fields: 'Album name : Angela's Bday' and 'timeline :'. The 'timeline' field is a native OS-style scrolling date picker. It has a header with '< SEP 2025 >' and a grid of dates. The date '8' is selected and highlighted. Below the date picker are two sections: 'PEOPLE' with four empty circles and a pencil icon, and 'PHOTOS' with a 3x4 grid of empty square boxes. Each box in the 'PHOTOS' grid has a small circular icon in its top-left corner (minus for removal, plus for addition). At the bottom of the screen is a navigation bar with three tabs: 'GALLERY', 'ALBUMS', and 'ME'.

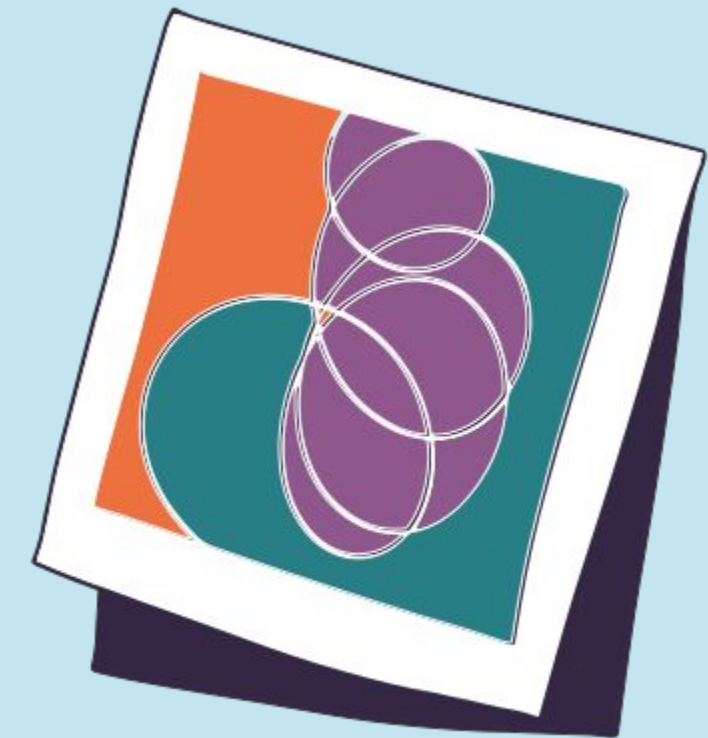
Benefits of Change #2

Evidence from Testing:

- Our testing data and TA/Peer feedback explicitly noted "confusion around timeline sliders" when adjusting the album start/end times (Severity 2 incident). Users didn't intuitively know how to interact with the plain boxes.

Efficiency (Time on Task):

- Utilizing familiar, standard platform UI components allows users to instantly know how to interact without guessing. This significantly reduces cognitive load, prevents input errors, and directly improves our Learnability metric.



Major Change 3 - Direct Avatar Editing

Previously

Users attempted to tap avatars directly but were required to click a separate edit icon, leading to misclicks and confusion

The 'EDIT ALBUM' screen in the previous design had a title bar with 'EDIT ALBUM' and a hamburger menu icon. Below the title bar, there were three input fields: 'Album name' (containing 'Angela's Bday'), 'Album start' (containing 'APRIL 11' and '5 PM'), and 'Album end' (containing 'APRIL 11' and '9 PM'). Below these fields, there was a section titled 'PEOPLE' with four circular avatars and a separate circular icon containing a pencil. Below the 'PEOPLE' section, there was a section titled 'PHOTOS' with a 3x4 grid of photo thumbnails. Each thumbnail had a small minus icon in the top-left corner. At the bottom of the 'PHOTOS' section, there was a 'SEND >' button. At the very bottom of the screen, there was a navigation bar with three tabs: 'GALLERY', 'ALBUMS', and 'ME'.

Now

Avatars are directly tappable to remove participants, aligning with user expectations and reducing interaction steps

The 'EDIT ALBUM' screen in the current design has a title bar with 'EDIT ALBUM' and a hamburger menu icon. Below the title bar, there are three input fields: 'Album name' (containing 'Angela's Bday'), 'Album start' (containing 'APRIL 11' and '5 PM'), and 'Album end' (containing 'APRIL 11' and '9 PM'). Below these fields, there is a section titled 'PEOPLE' with four circular avatars, each containing a small 'x' icon in the top-right corner, and a circular icon containing a plus sign. Below the 'PEOPLE' section, there is a section titled 'PHOTOS' with a 3x4 grid of photo thumbnails. Each thumbnail has a small minus icon in the top-left corner. At the bottom of the 'PHOTOS' section, there is a 'SEND >' button. At the very bottom of the screen, there is a navigation bar with three tabs: 'GALLERY', 'ALBUMS', and 'ME'.

Benefits of Change #3

Evidence from Testing:

- During usability testing, 2 out of 3 users attempted to tap on the avatar icons directly when trying to remove someone from an album. However, the original design required users to click a small pencil icon first, leading to navigation errors and hesitation.

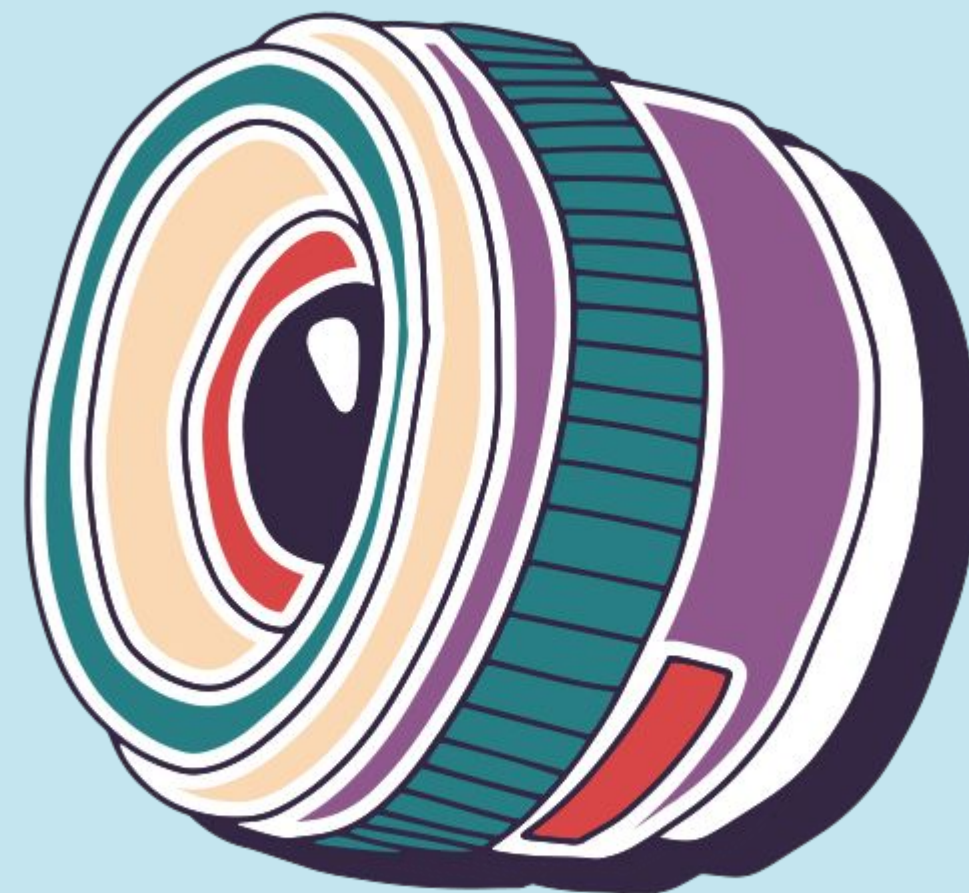


Efficiency (Time on Task):

- Removing the extra “Edit” step eliminates one interaction layer, shortening the moderate task flow. Fewer steps and clearer affordances are expected to reduce time on task and minimize hesitation, particularly for first-time users.

05

medium-fi task flows & prototype



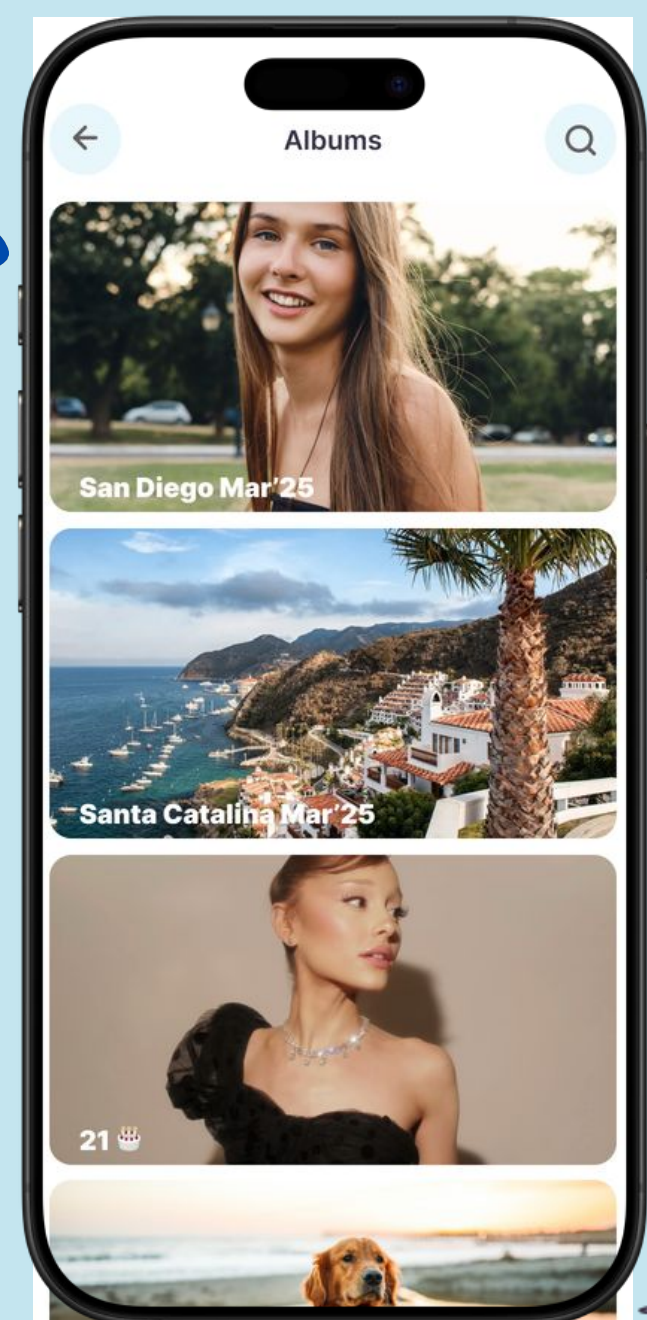
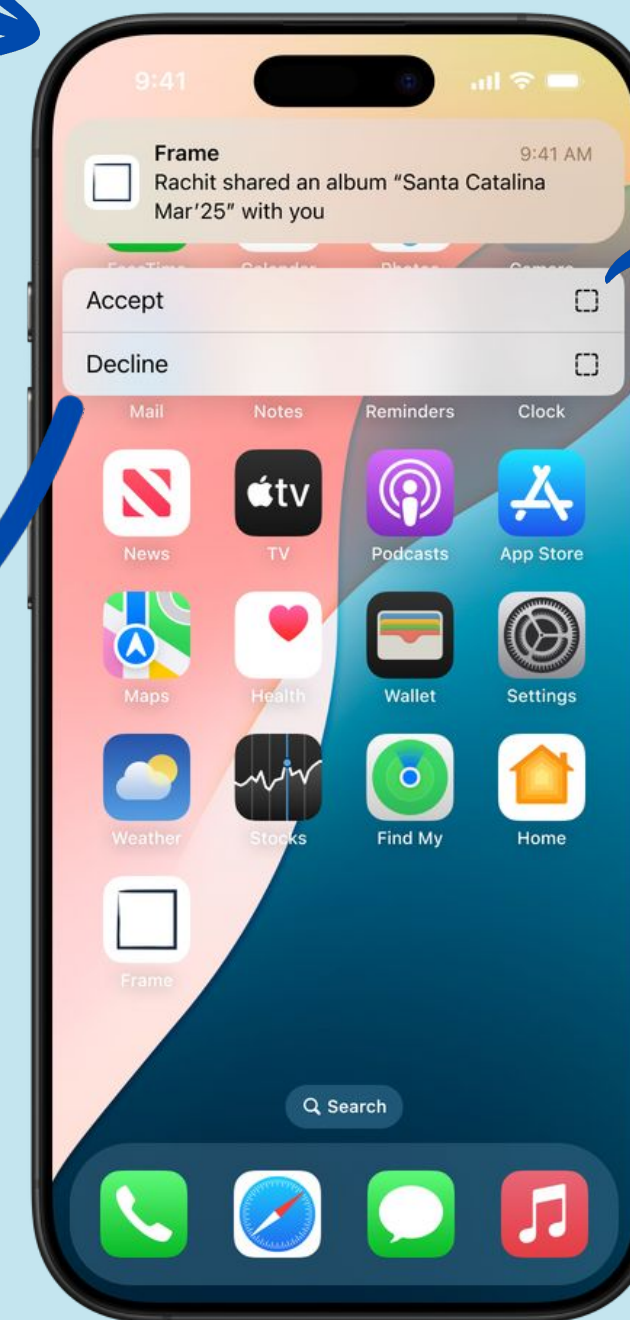
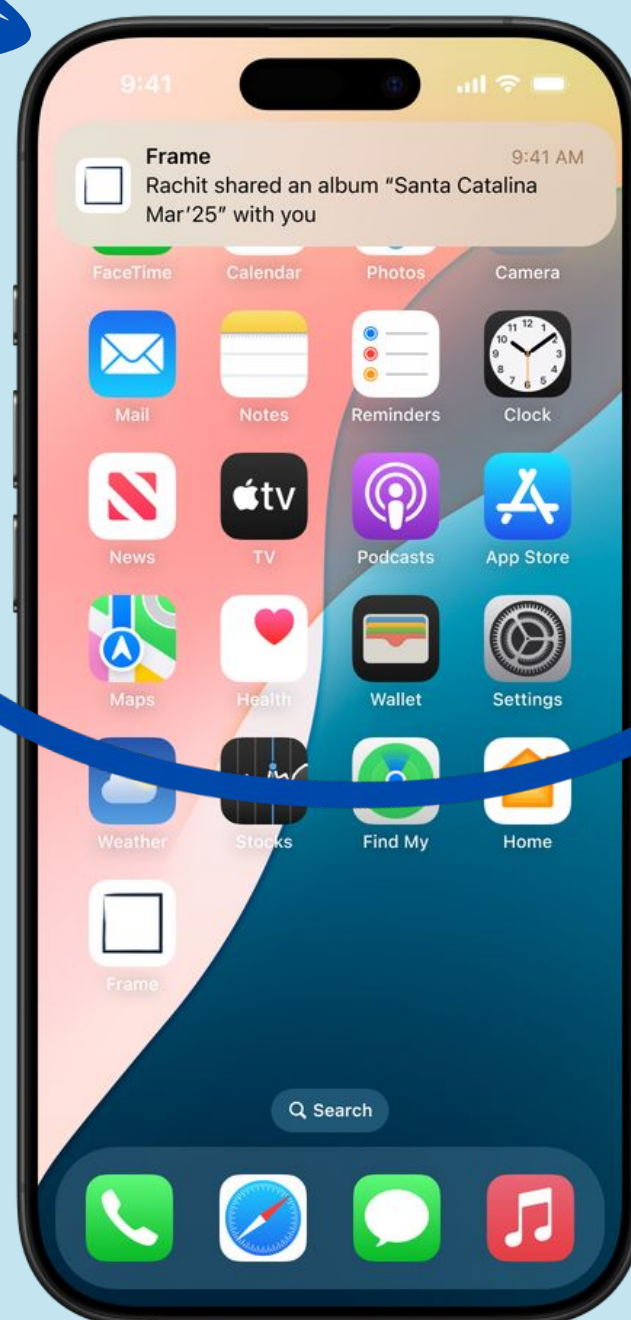
simple

User receives shared album and accepts or decline it

user gets a notification that an album is shared with them

user holds on the notification to see options

if accepted, adds the album to their albums page



if rejected, the notification goes away

moderate

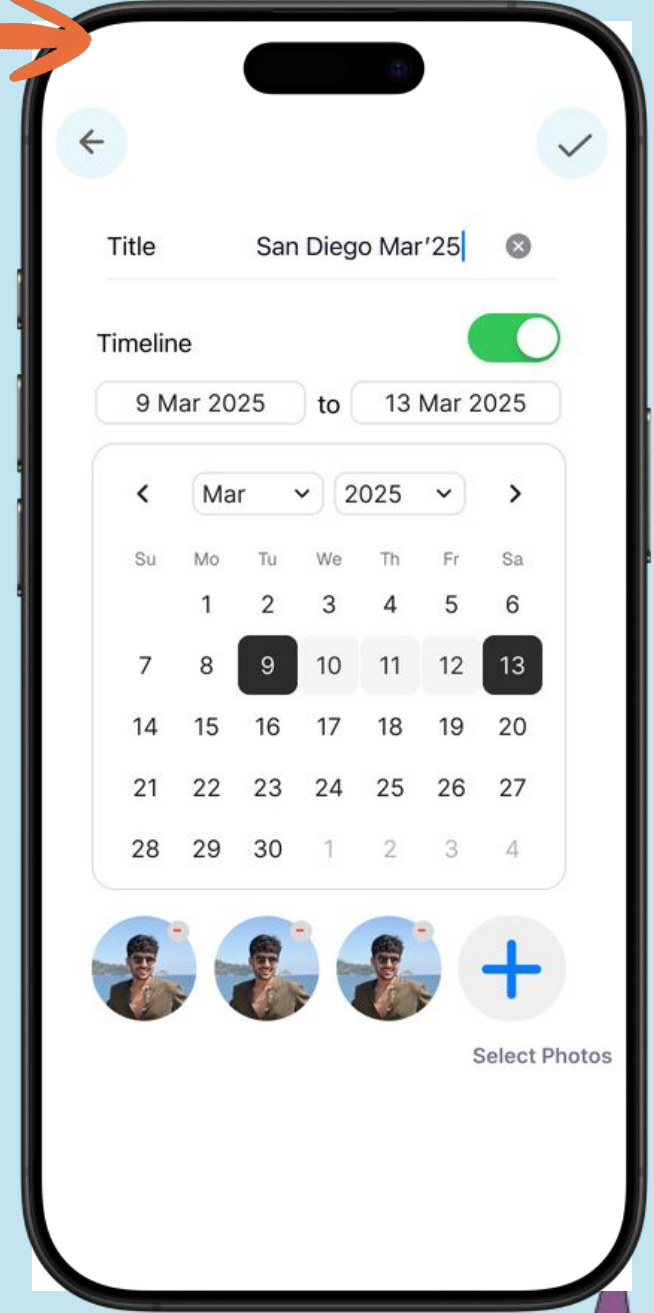
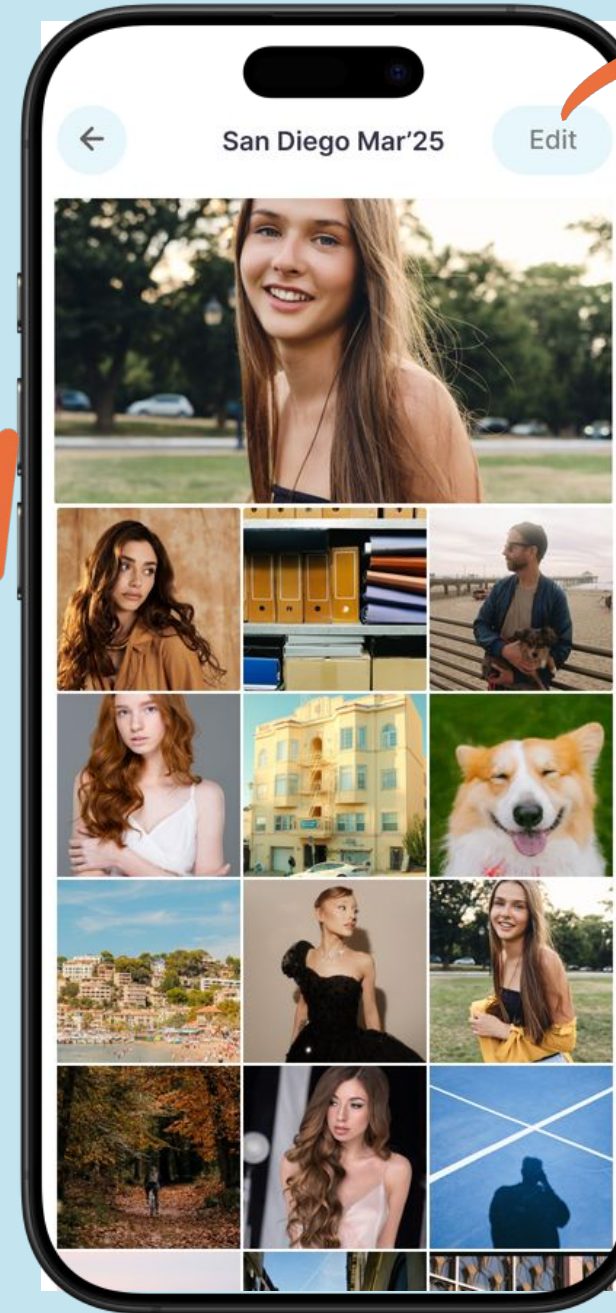
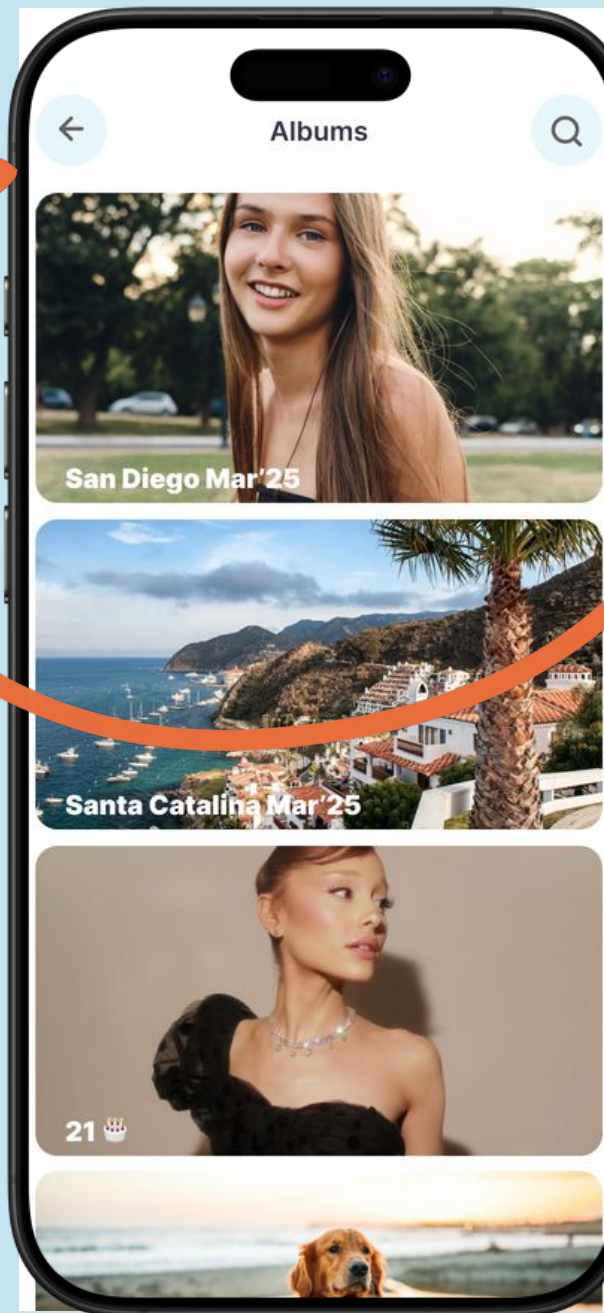
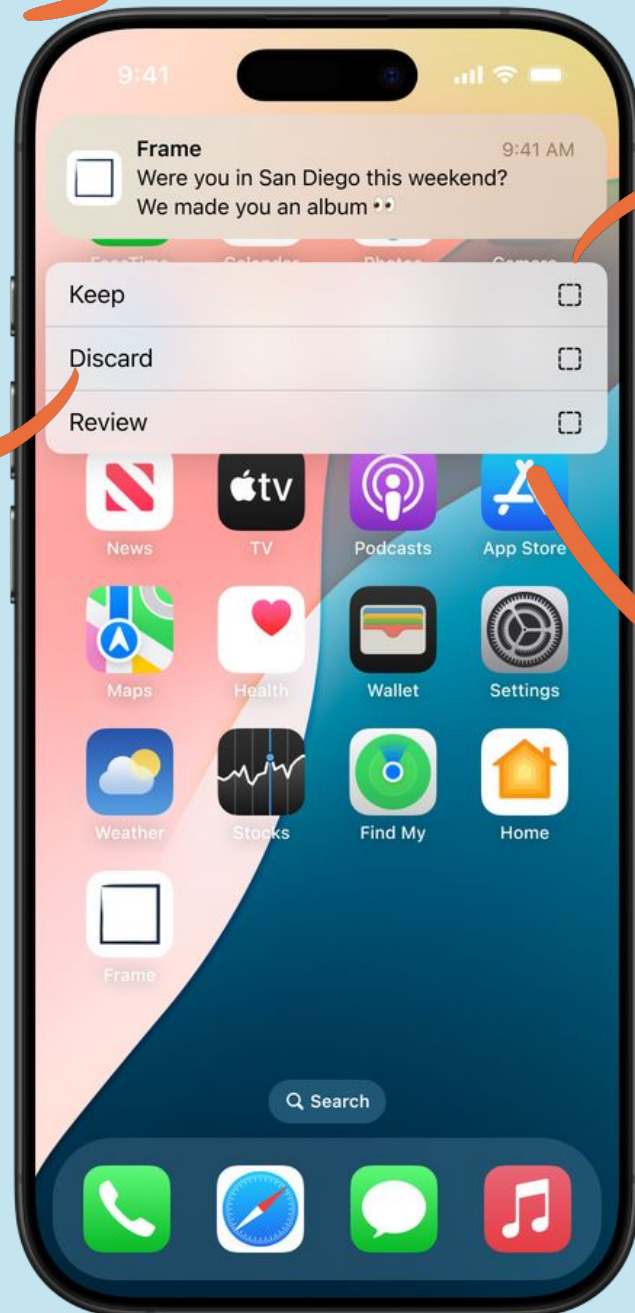
User discovers automatically curated albums by frame

user gets notification
when an automated
album is created

user holds on the notification to
see options

if Keep, adds the album to their
albums page

user can edit name, timeline,
people who have access to it and so
on



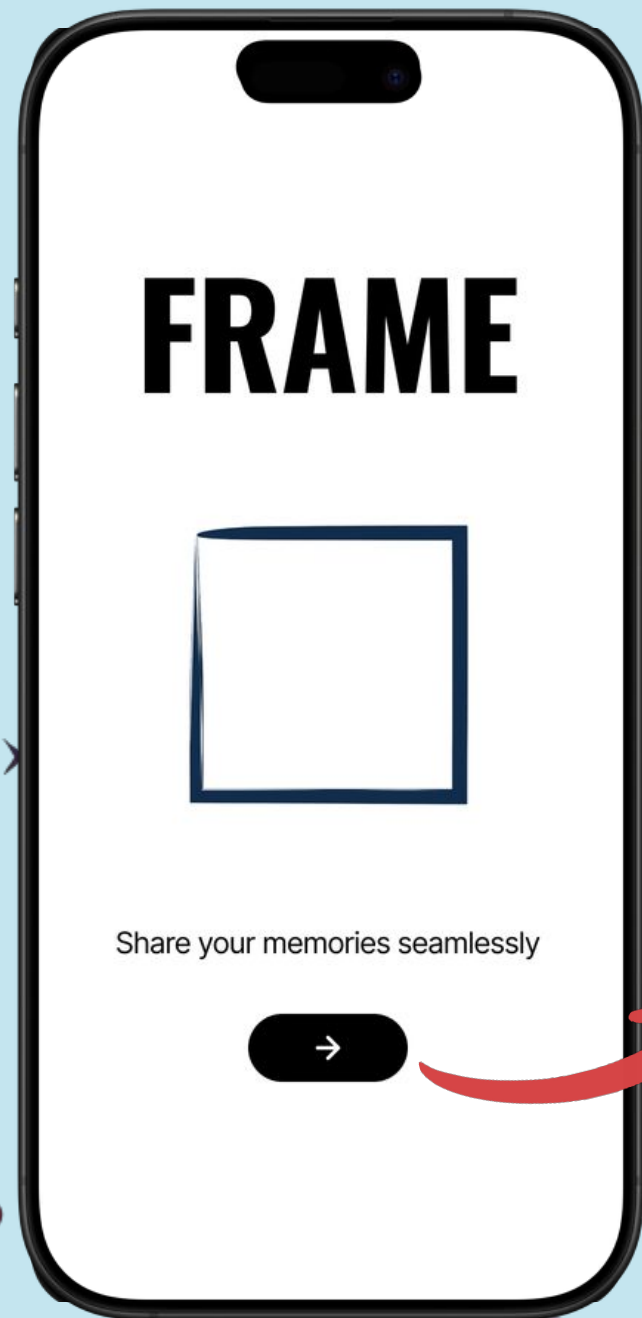
if Discard, notification goes away
and album is discarded

if Review, user can access and
edit the album

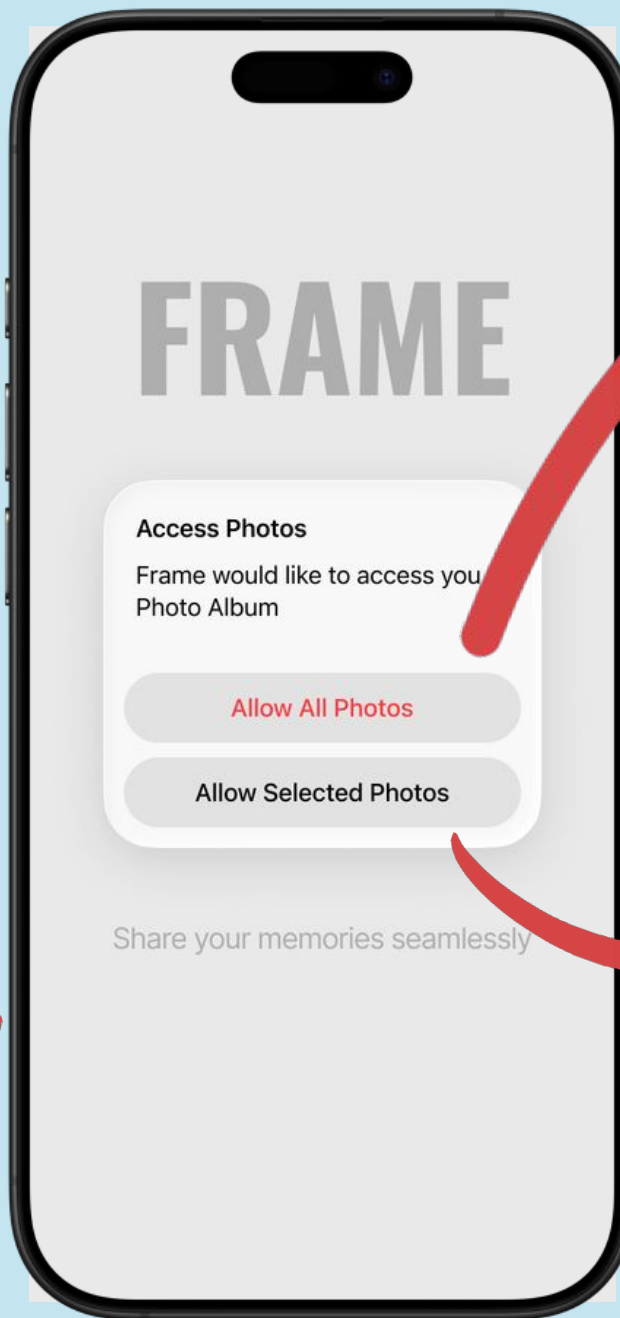
complex

User imports all or selected photos from his gallery into the app

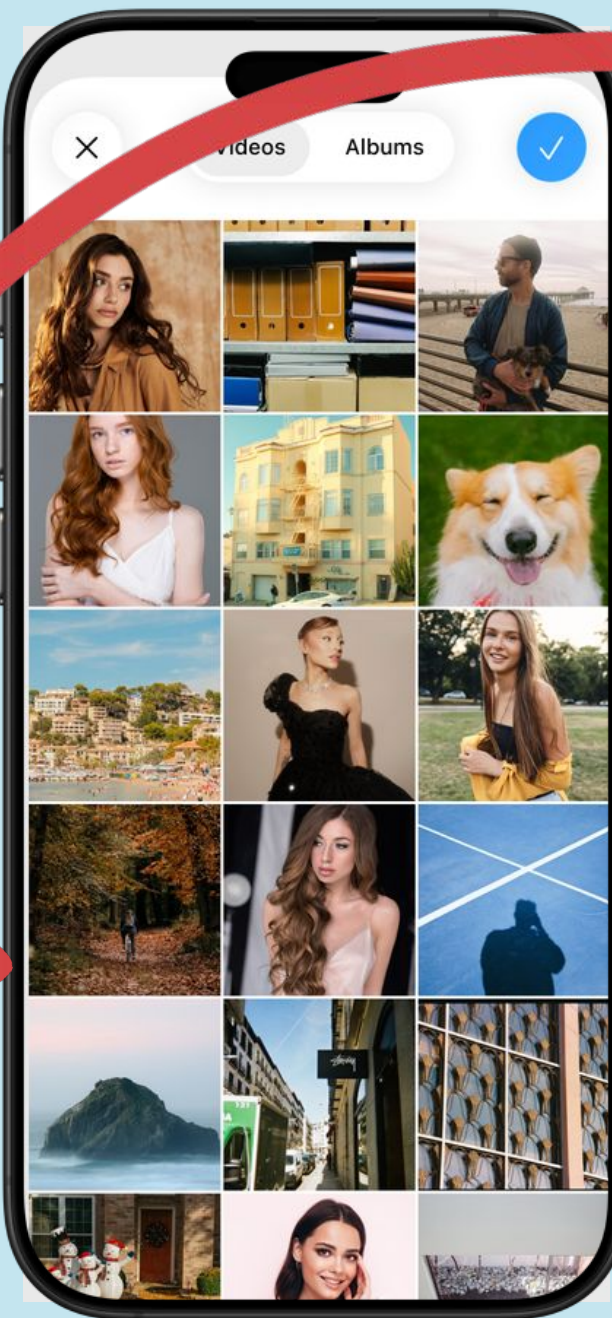
user opens the app from the home page during setup



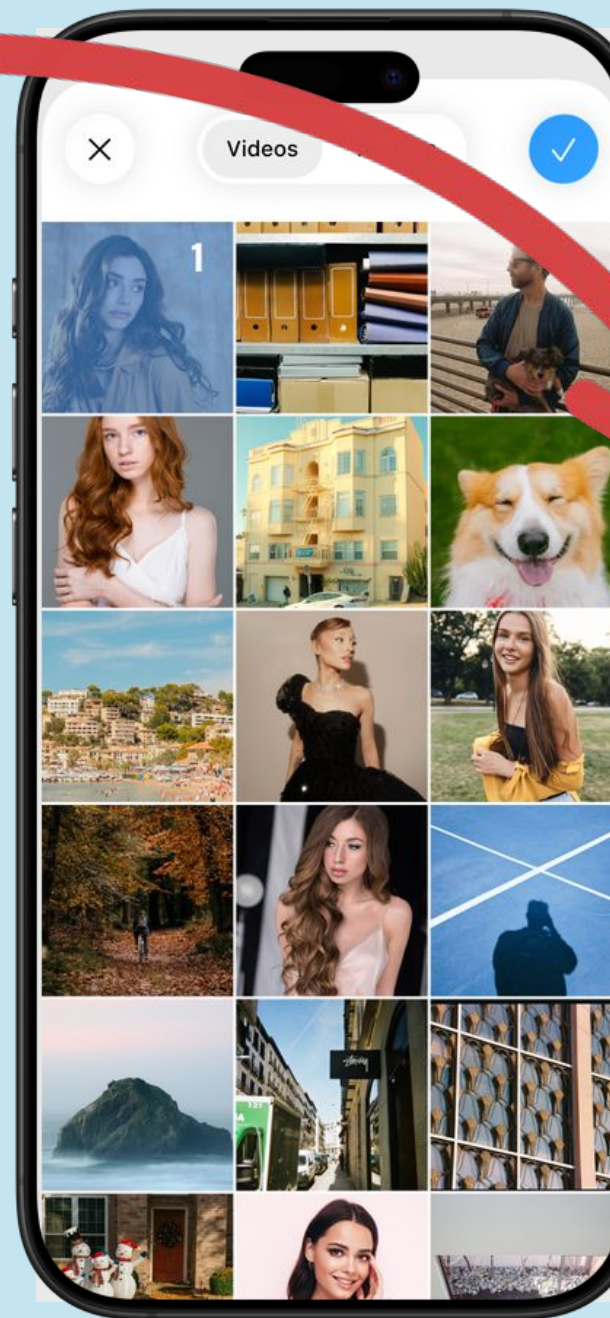
user gets a pop up notification to allow frame to get access to photos



If user only wants to allow selected photos, then they are prompted with their gallery to chose which ones

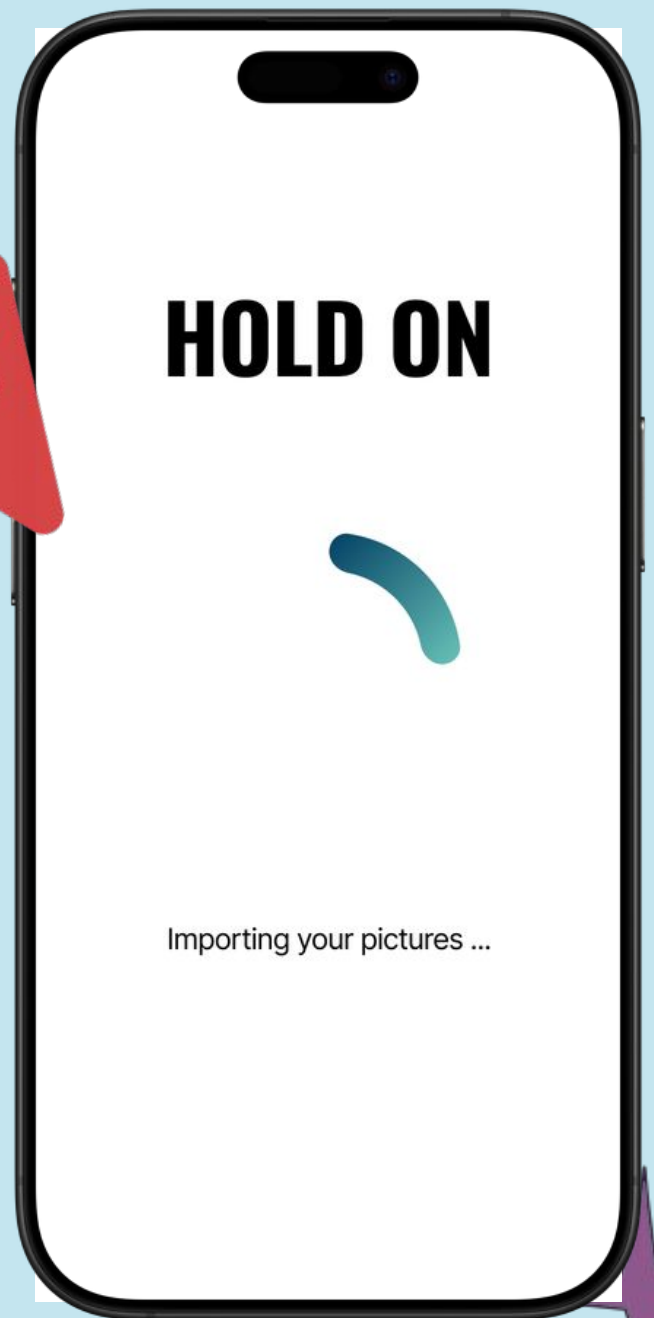


if user allows all access, frame starts importing all of the photos from the gallery



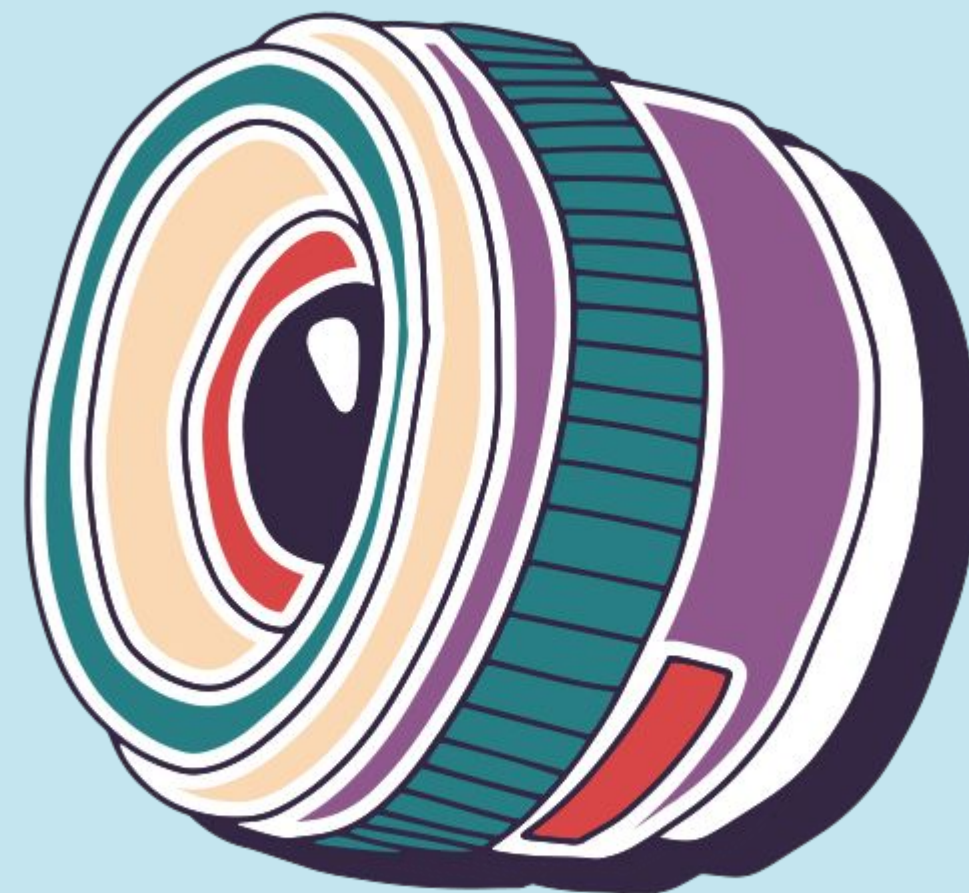
users select the photos

after selection, frame starts importing the selected photos

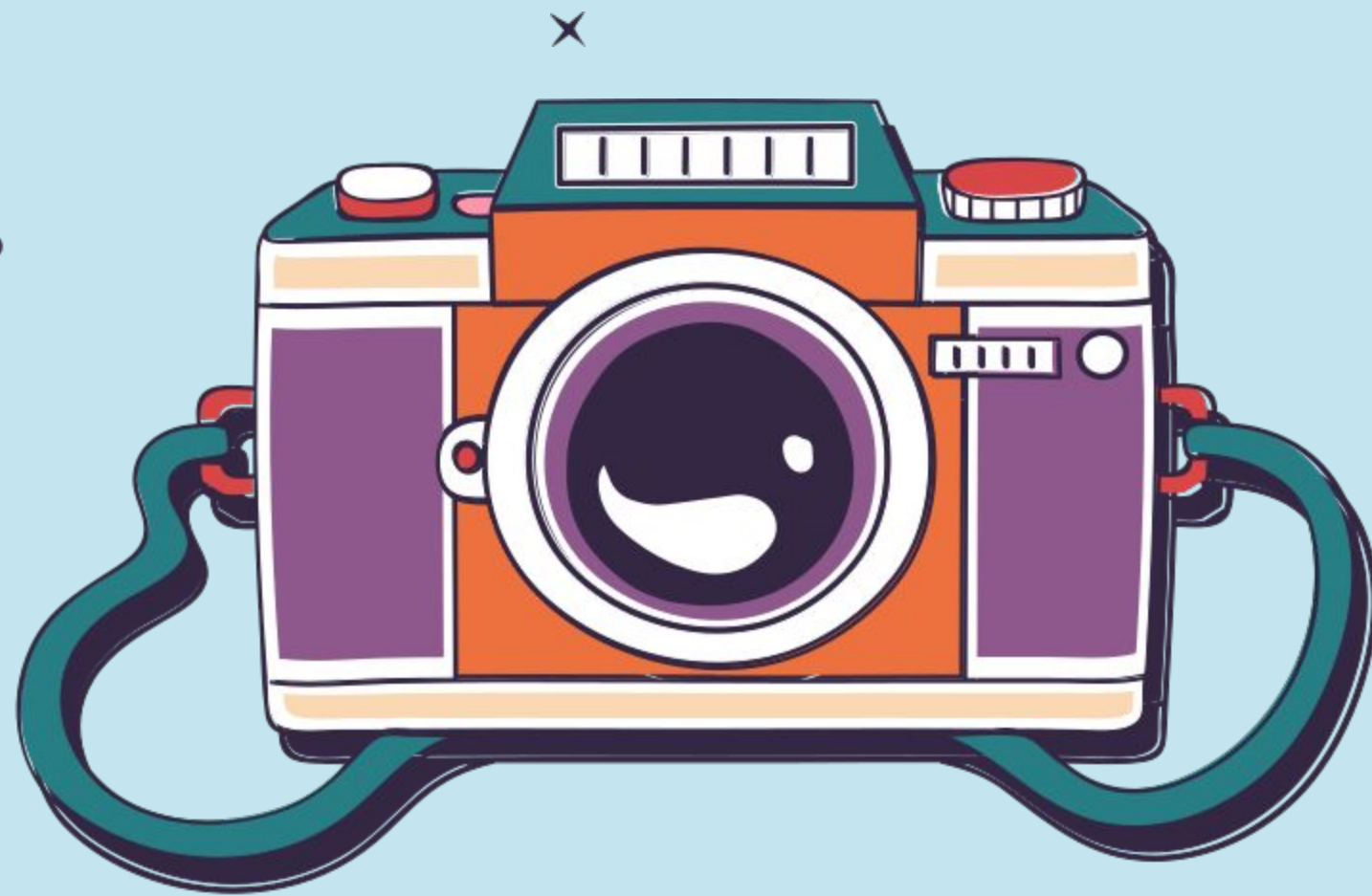




prototype implementation



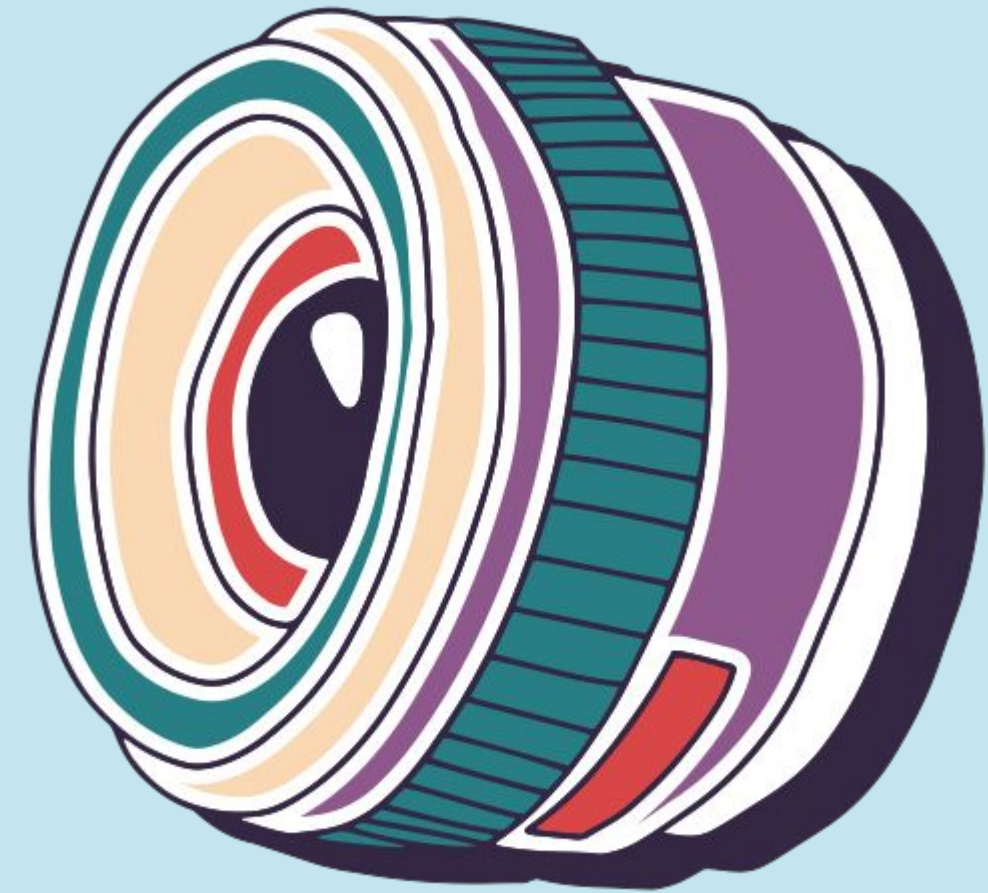
limitations



- Users cannot test whether albums are accurately generated
- Users cannot test whether the “right” people are included
- Users cannot truly test privacy boundaries or experience real data filtering behavior
- We cannot evaluate how real upload speeds, sync delays, or processing times would affect perceived efficiency

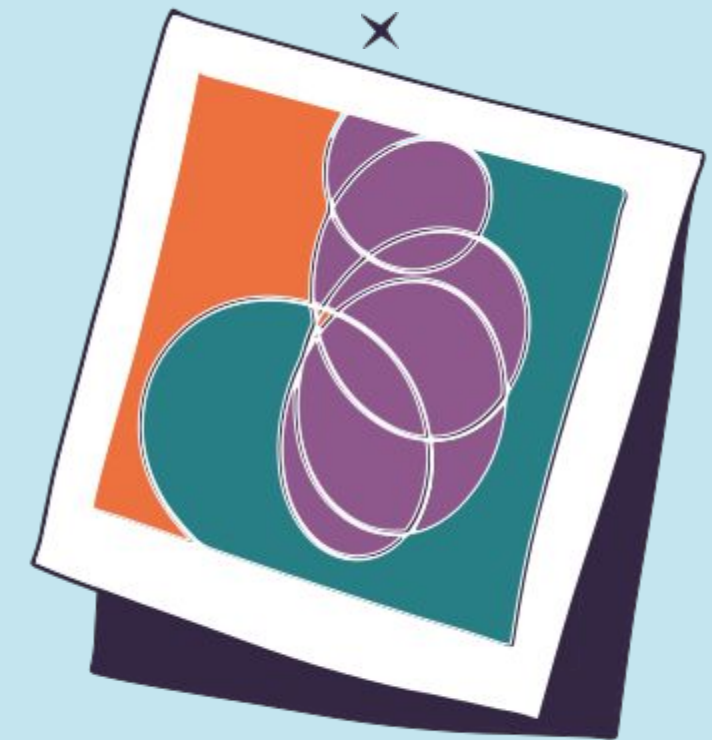


prototype features



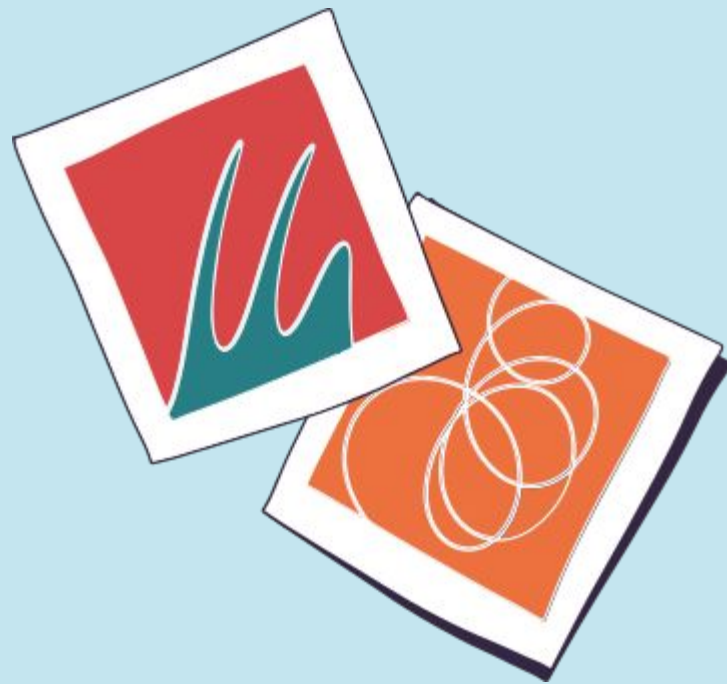
wizard of oz & some hardcoded features

- All the gallery content is hard-coded, so users always see the same pre-loaded set of “photos” rather than importing their own real camera roll images.
- The system always generates the same album (same cover photo, same title, same people) instead of actually detecting an event or grouping photos intelligently.
- Notifications are hard-coded per task flow, so users only receive one fixed notification type (e.g., “Were you in Chicago this weekend?”) rather than varied notifications across different events, people, or dates.
- The privacy/permission pop-up uses a fixed selection path, so even if “Select Photos” is chosen, the prototype only allows selecting one photo



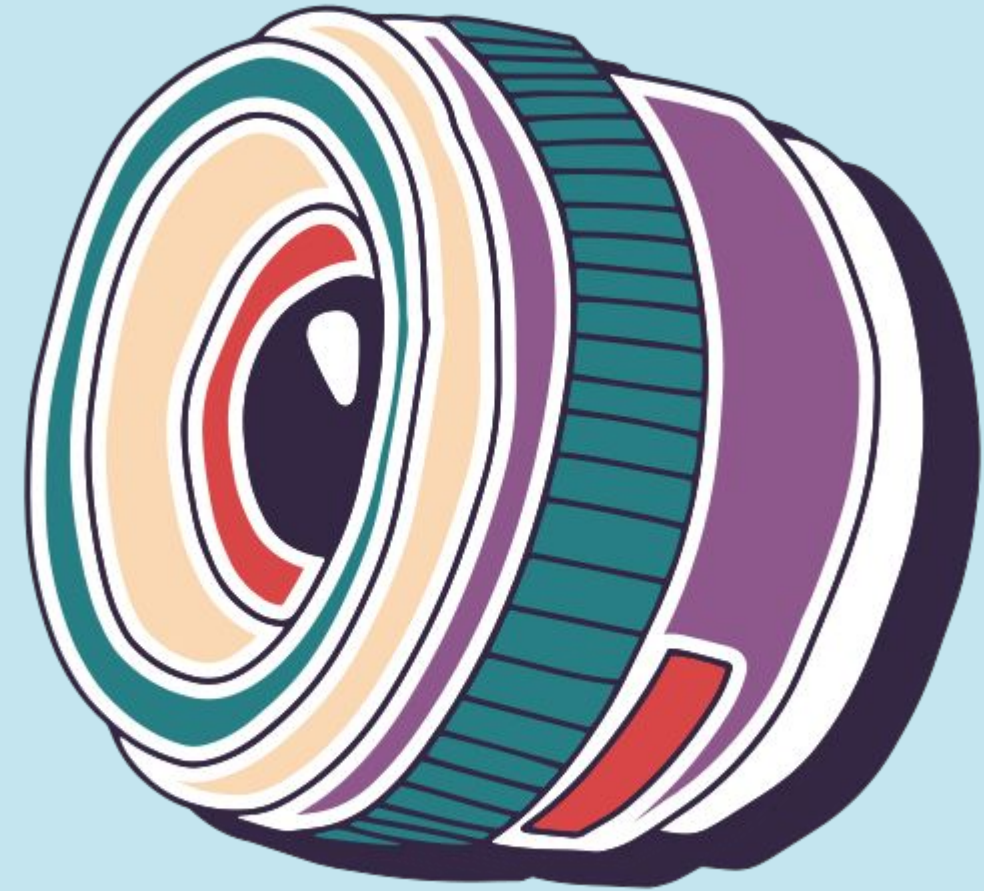
impacts

- Because the gallery content is hard-coded, users are not interacting with their own real photos, which may reduce emotional attachment and limit our ability to evaluate trust, privacy sensitivity, and real-world hesitation.
- Since the same album is always generated with the same people and cover photo, we cannot evaluate how users respond to incorrect AI groupings or missing photos, which may artificially inflate perceived learnability and efficiency.
- Hard-coded notifications prevent us from testing how users interpret varied event prompts, so we cannot measure whether confusion increases when event names, people, or contexts change.
- The fixed permission flow (only allowing one photo selection) simplifies the complexity of real privacy decisions, potentially underestimating the mental effort required when users must select multiple photos or manage large libraries.





appendix

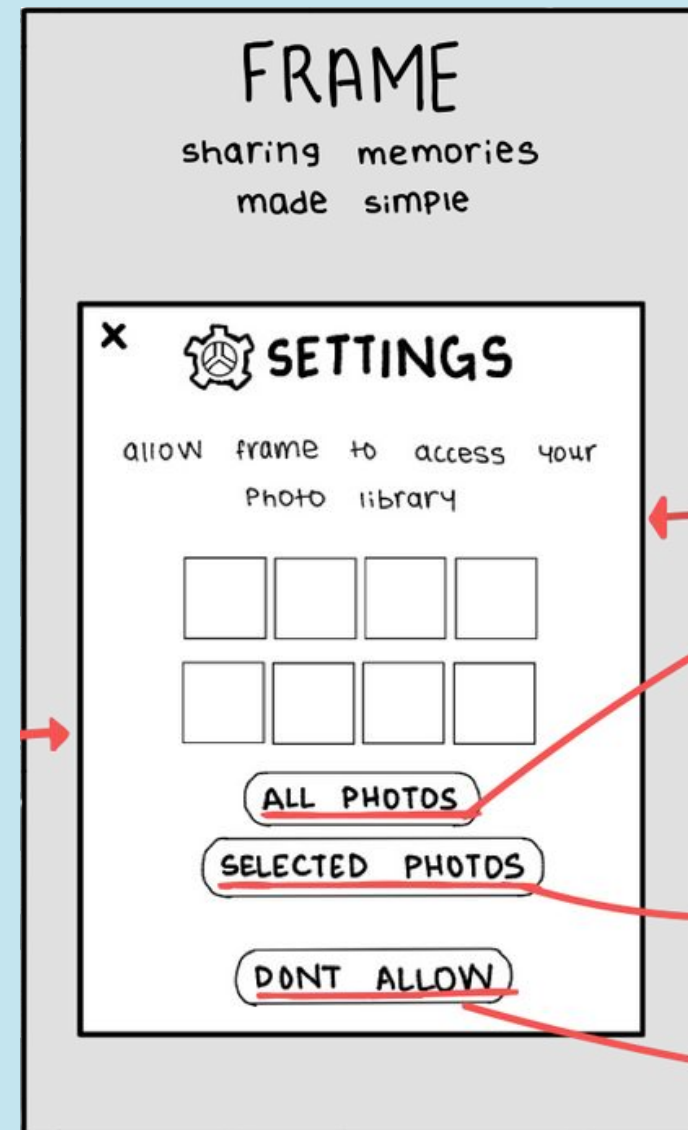




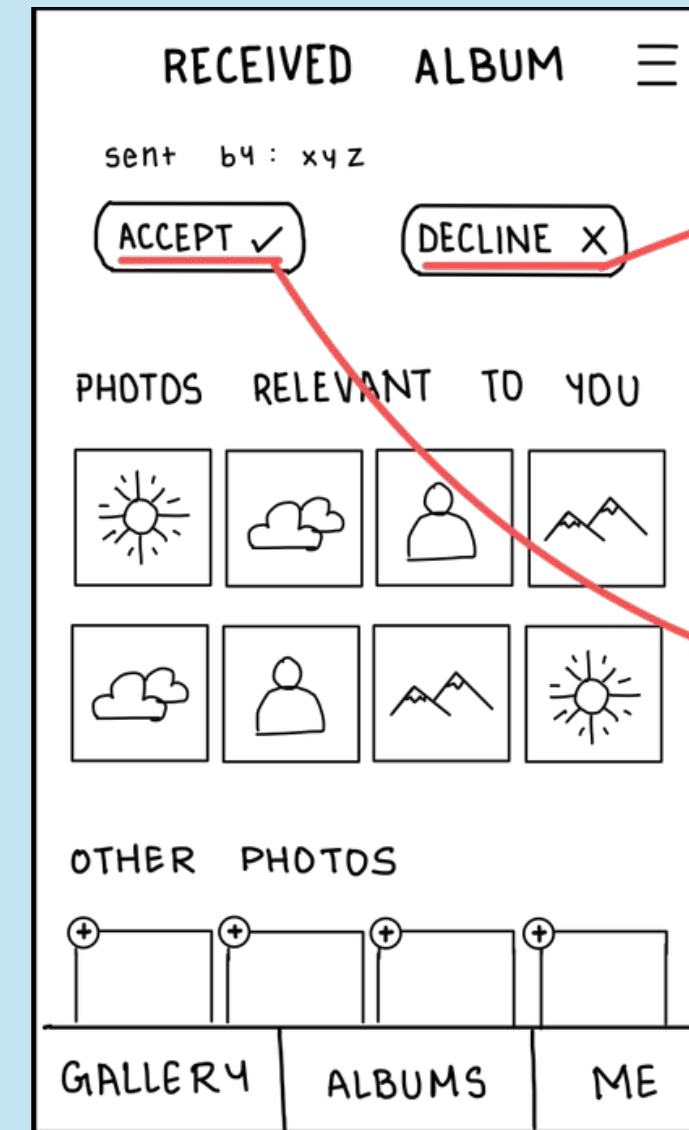
[Figma Link](#) + [Readme Link](#)



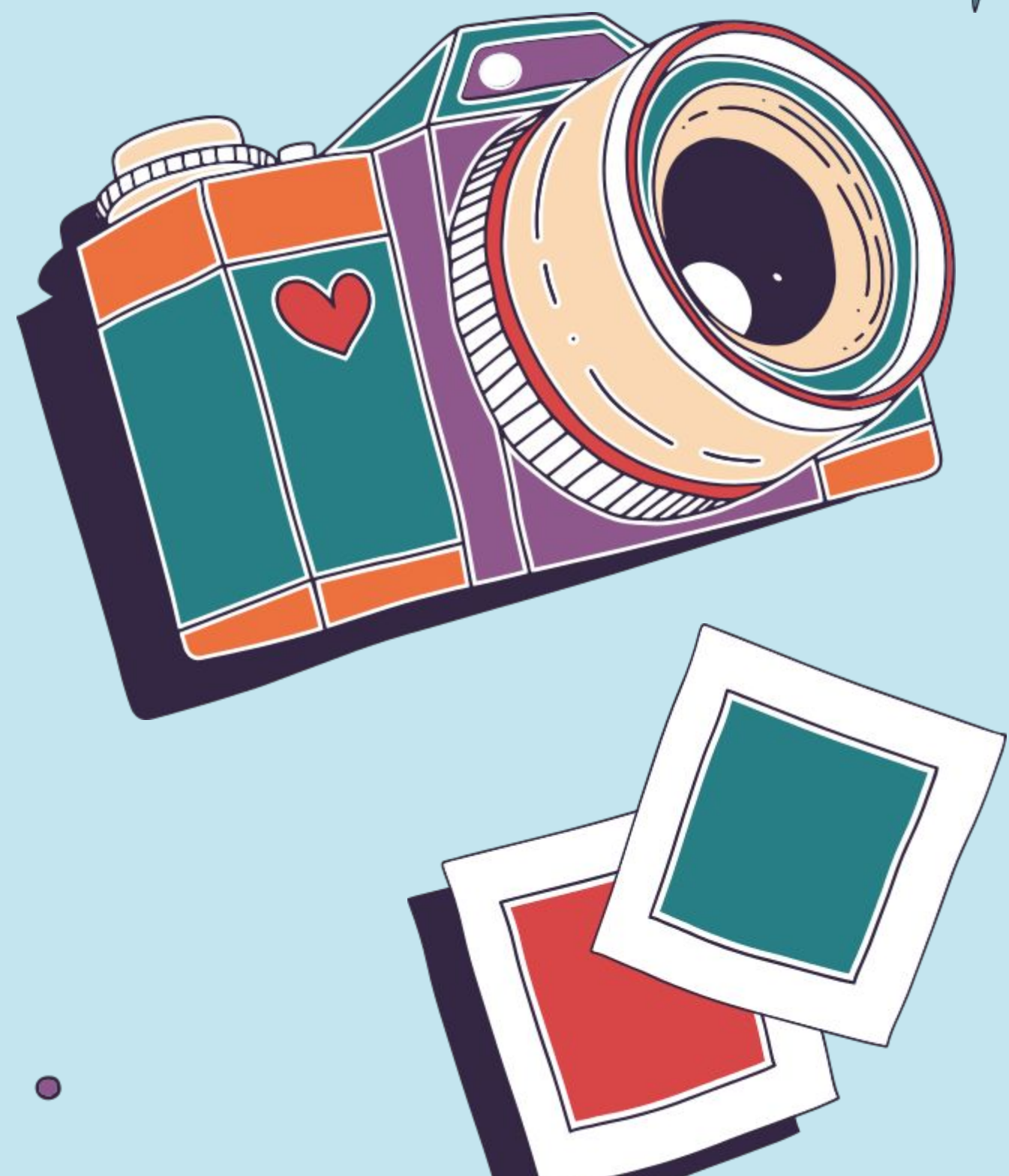
Other Changes



Native OS-permission pop up doesnt have a don't allow option



We don't need separate received albums page to accept or decline the album, it can be done just by holding on the notification



thank you!